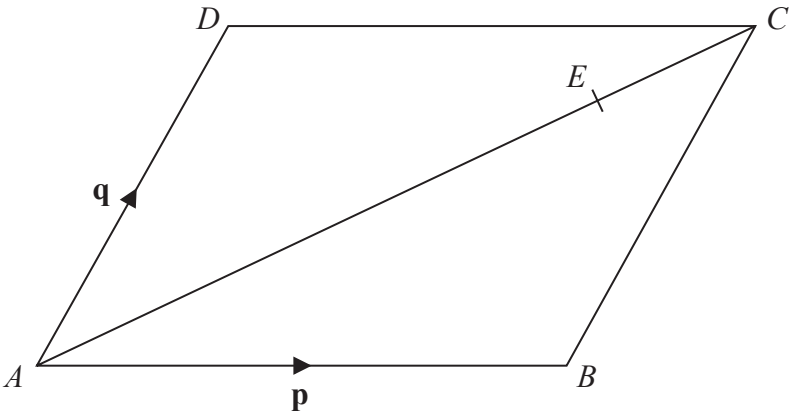
14



$ABCD$ is a parallelogram.

$\overrightarrow{AB} = \mathbf{p}$

$\overrightarrow{AD} = \mathbf{q}$

The point E lies on AC such that $AE : EC = 4 : 1$

Find $\overrightarrow{DE}$ in terms of **p** and **q**.
Give your answer in its simplest form.

$AC = AB + BC = p + q$

1 mark
Diagonal vector.

$AE = \frac{4}{5}(p + q)$

1 mark
AE is 4 of the 5 parts.

$DE = DA + AE = -q + \frac{4}{5}(p + q)$

1 mark
Travel D → A → E.

$DE = \frac{4}{5}p - \frac{1}{5}q$

1 mark
Collect p and q.

(Total for Question 14 is 4 marks)

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